
Algorithm Performance Analyzer

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System Report

1. Introduction

This project provides an empirical and visual way to analyze algorithm performance beyond theoretical Big-O notation.

2. Problem Statement

Traditional analysis focuses on theory only. This system measures real execution performance across multiple scenarios.

3. System Architecture

The system consists of a Flutter frontend, Node.js backend, and an analysis engine responsible for benchmarking and complexity estimation.

4. Features

- Auto & Manual input modes
- Real-time performance analysis
- Visualization using charts
- PDF export
- Complexity detection ($O(n)$, $O(n^2)$, $O(\log n)$, $O(n \log n)$)

5. Analysis Engine

Includes AST parsing, benchmark execution, slope calculation (k values), and classification into best, average, and worst cases.

6. Data Flow

User inputs code → sent to backend → analyzed → benchmark executed → results returned → visualized in frontend.

7. Technologies Used

Frontend: Flutter, fl_chart

Backend: Node.js, Express

Others: PDF generation, Screenshot tools

8. Conclusion

The project enhances understanding of algorithm performance using real-world measurements and visual tools.

9. System Design Diagram

The following diagram represents the system architecture:

